

Section : Biology

Q.1 The rate of biomass production is expressed in terms of _____ to compare the productivity of different ecosystems.

1. $\text{kg}^{-1} \text{yr}^{-2}$ or $(\text{kcal m}^{-1}) \text{yr}^{-2}$
2. $\text{gm}^{-2} \text{yr}^{-2}$ or $(\text{kcal m}^{-2}) \text{yr}^{-2}$
3. $\text{kg}^{-1} \text{yr}^{-1}$ or $(\text{kcal m}^{-1}) \text{yr}^{-1}$
4. $\text{gm}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$

Options 1. 1

2. 2

3. 3

4. 4

Q.2 If there are 4 states, represented as L, M, N, O where L is located near the equator, and others are located at 80°N , 71°N , 41°N , respectively. Which one of the following will have least number of bird species?

1. L
2. M
3. N
4. O

Options 1. 1

2. 2

3. 3

4. 4

Q.3 The hormone releasing IUDs function by-

1. Make the cervix unsuitable for implantation and the uterus hostile to the sperms
2. Make the uterus unsuitable for fertilization and the cervix hostile to the sperms
3. Make the fallopian tube unsuitable for fertilization and the cervix hostile to the sperms
4. Make the uterus unsuitable for implantation and the cervix hostile to the sperms

Options 1. 1

2. 2

3. 3

4. 4

Q.4 Which of the following factors can influence Hardy-Weinberg equilibrium?

- (A) Natural Selection
- (B) Genetic recombination
- (C) Gene flow
- (D) Mutation

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B), (C), (D)
3. (A), (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.5 Which of the following is not the alien invasive species?

1. *Parthenium*
2. *Clarias gariepinus*
3. *Quagga*
4. *Lantana*

Options 1. 1

2. 2

3. 3

Q.6 The Medical Termination of Pregnancy (Amendment) Act, 2017 was enacted by the government of India with the intension of_____.

1. Reducing the incidence of illegal abortion and consequent paternal mortality and morbidity
2. Reducing the incidence of legal delivery and consequent maternal mortality and morbidity
3. Reducing the incidence of legal abortion and consequent maternal mortality and morbidity
4. Reducing the incidence of illegal abortion and consequent maternal mortality and morbidity

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Which of the following biofertilisers are free living in the soil?

- (A) *Anabaena*
- (B) *Azospirillum*
- (C) *Rhizobium*
- (D) *Azotobacter*

Choose the correct answer from the options given below:

1. (B) and (C) only
2. (A) and (B) only
3. (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.8 The role of ethidium bromide in gel electrophoresis is:

1. To separate the DNA fragments in gel electrophoresis
2. To speed up the electrophoresis
3. To neutralize DNA fragments
4. To stain DNA which helps in identifying the desired DNA fragment under UV exposure

Options 1. 1

2. 2
3. 3
4. 4

Q.9 If primary producers are exposed to 1,000,000 Joules of sunlight, the amount of energy transferred to tertiary consumers will be:

1. 100 J
2. 1000 J
3. 10000 J
4. 10 J

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Match List-I with List-II

List-I

List-II

Conditions

Explanations

(A) Apocarpous

(I) Gynoecium consists of a single pistil

(B) Syncarpous

(II) Gynoecium consists of a more than one pistil

(C) Multicarpellary

(III) More than one pistils fused together

(D) Monocarpellary

(IV) More than one free pistils

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
2. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
3. (A) - (III), (B) - (IV), (C) - (I), (D) - (II)
4. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

Options 1. 1

2. 2
3. 3
4. 4

Q.11 In a typical test cross-

1. An organism showing a recessive phenotype is crossed with the dominant parent
2. An organism showing a dominant phenotype is crossed with the recessive parent
3. An organism showing a dominant genotype is crossed with the dominant parent
4. An organism showing a recessive genotype is crossed with the recessive parent

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Which of the following enzymes are used to clarify the packed juices?

- (A) Pectinases
- (B) Kinases
- (C) Proteases
- (D) Nucleases

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A) and (C) only
3. (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.13 The correct sequence of formation of following structures in the developing embryo is -

- (A). Heart
- (B). Hair on head
- (C). Limbs and digits
- (D). Eyelashes

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Gause's 'Competitive Exclusion Principle' states that -

1. Two closely related species competing for the different resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.
2. Two distantly related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.
3. Two closely related species competing for the same resources cannot co-exist indefinitely and the competitively inferior one will be eliminated eventually.
4. Two closely related species competing for the same resources can co-exist indefinitely and the competitively inferior one will be eliminated

eventually.

Options 1. 1

2. 2
3. 3
4. 4

Q.15 Arrange the following processes of recombinant DNA technology in a correct sequence:

- (A). Amplification of gene by using PCR
- (B). Isolation of DNA and cutting of DNA at specific location
- (C). Obtaining the desired foreign gene product
- (D). Insertion of recombinant DNA into the host

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (A), (C), (B), (D)
3. (B), (A), (D), (C)
4. (C), (B), (D), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.16 Arrange the given organisms involved in a food chain in correct sequence ?

- (A) Fish
- (B) Phytoplankton
- (C) Man
- (D) Zooplankton

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)
2. (B), (D), (A), (C)
3. (C), (D), (A), (B)
4. (D), (C), (B), (A)

Options 1. 1

2. 2
3. 3
4. 4

Q.17 If sewage water is added to river water, it will:

1. Increase the BOD of sewage water
2. Increase the BOD of river water
3. Decrease the BOD of river water
4. Not affect the BOD of river water and sewage water

Options 1. 1

2. 2
3. 3
4. 4

Q.18 If 4 individuals in a laboratory population of 40 fruitflies died during a specific time interval of one week. What will be the death rate of the population during that period?

1. 1.0 individual per fruitfly per week
2. 4.0 individual per fruitfly per week
3. 0.4 individual per fruitfly per week
4. 0.1 individual per fruitfly per week

Options 1. 1

2. 2
3. 3
4. 4

Q.19 Which one of the following pairs may indicate common ancestry?

1. Sweet potato - potato
2. Octopus eye - human eye
3. Forelimbs of man - forelimbs of whale

4. Wings of butterfly - wings of birds

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Which one of the following events is true about the *lac* operon in the absence of an inducer?

1. The repressor is inactivated

2. Repressor binds to the operator and prevents RNA polymerase from transcribing the operon

3. Beta-galactosidase, transacetylase and permease will be produced

4. RNA polymerase have access to the promoter and transcription proceeds

Options 1.

1

2. 2

3. 3

4. 4

Q.21 Match List-I with List-II

List-I

List-II

Type of pollinations

Features

(A) Autogamy (I) Flowers do not open

(B) Geitonogamy (II) Transfer of pollen grains from anther to stigma of different plant

(C) Xenogamy (III) Transfer of pollen grains from anther to stigma of another flower of same plant

(D) Cleistogamy (IV) Transfer of pollen grains from anther to stigma of the same flower

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

2. (A) - (II), (B) - (I), (C) - (IV), (D) - (III)

3. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

4. (A) - (III), (B) - (II), (C) - (I), (D) - (IV)

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Fertilization in human female takes place in -

1. Uterus

2. Fimbriae portion of fallopian tube

3. Ampullary region of fallopian tube

4. Isthmus region of fallopian tube

Options 1. 1

2. 2

3. 3

4. 4

Q.23 The relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, freshwater fishes) can be represented by the equation :

1. $\log Z = \log C + S \log A$

2. $\log A = \log C + Z \log S$

3. $\log C = \log S + Z \log A$

4. $\log S = \log C + Z \log A$

Options 1. 1

2. 2

3. 3

Q.24 If the sequence of one strand of DNA is written as: 5'-ATCGTGCTATGAGCATGC-3'.

The sequence of its complementary strands will be?

1. 5'-TAGCACGATACTCGTACG-3'
2. 3'-TAGCACGATACTCGTACG-5'
3. 3'-UAGCACGUAACCTCGTACG-5'
4. 3'-UAGCACGAUACUCGUACG-5'

Options 1. 1

2. 2
3. 3
4. 4

Q.25 ZIFT is an assisted reproductive technique in which:

1. The early embryos, more than 8 blastomeres are transferred into the fallopian tube
2. The zygote with upto 8 blastomeres are transferred into the uterus
3. The zygote or early embryos with upto 8 blastomeres are transferred into the fallopian tube
4. The ovum is transferred from the donor into the fallopian tube of another female

Options 1. 1

2. 2
3. 3
4. 4

Q.26 Which one of following hormones is not produced during the pregnancy in women?

1. hCG
2. Relaxin
3. Androgens
4. hPL

Options 1. 1

2. 2
3. 3
4. 4

Q.27 The loss of biodiversity in a region may not lead to

1. Decline in plant production
2. Lowered resistance to environmental perturbations such as drought
3. Decline in plant production but increase in animal populations
4. Increased variability in certain ecosystem processes such as plant productivity, water use, and pest and disease cycles

Options 1. 1

2. 2
3. 3
4. 4

Q.28 A chromosomal disorder caused due to loss of one X chromosome in human female is:

1. Down's syndrome
2. Klinefelter's syndrome
3. Turner's syndrome
4. Acquired immuno deficiency syndrome

Options 1. 1

2. 2
3. 3
4. 4

Q.29 Match List-I with List-II

List-I

List-II

Microbes	Products or uses
(A). <i>Trichoderma polysporum</i>	(I). Blood cholesterol lowering agent
(B). <i>Streptococcus</i>	(II). Citric acid
(C). <i>Aspergillus niger</i>	(III). Clot buster to remove blood clots
(D). <i>Monascus purpureus</i>	(IV). Cyclosporin A

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (IV), (B) - (I), (C) - (II), (D) - (III)
- (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

Options 1. 1

- 2
- 3
- 4

Q.30 The identifying features of the embryo of monocotyledons are:

- Coleoptile
- Root cap enclosed in coleorhiza
- Pericarp
- One cotyledon

Choose the correct answer from the options given below:

- (A), (B) and (C) only
- (A), (B) and (D) only
- (A), (B), (C) and (D)
- (B), (C) and (D) only

Options 1. 1

- 2
- 3
- 4

Q.31 Match List-I with List-II

List-I	List-II
Techniques	Features
(A) Bioprocess engineering	(I) Use of bio resources without proper authorisation
(B) Somatic hybridisation	(II) Correction of gene defect
(C) Gene therapy	(III) Fusion of protoplasts from two different organisms
(D) Biopiracy	(IV) Maintenance of sterile ambience for microbial growth

Choose the correct answer from the options given below:

- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

- 2
- 3
- 4

Q.32 Secretions of which structure is not essential for the maturation and motility of sperms.

- Epididymis
- Vas deferens
- Ureter
- Seminal vesicle

Options 1. 1

- 2
- 3

Q.33 Match List-I with List-II**List-I****Enzymes**

(A). DNA polymerase

(B). DNA ligase

(C). RNA polymerase

I

(D). RNA polymerase

II

List-II**Functions**

(I). Transcribes precursor of mRNA (hnRNA)

(II). Transcribes rRNAs

(III). Joins the DNA fragments

(IV). Catalyzes the polymerization of deoxynucleotides

the correct answer from the options given below:

1. (A)- (IV), (B) - (II), (C)- (I), (D)- (III)

2. (A)- (IV), (B)- (III), (C)- (II), (D)- (I)

3. (A)- (IV), (B) - (I), (C)- (II), (D)- (III)

4. (A)- (III), (B)- (IV), (C)- (I), (D)- (II)

Options 1. 1

2. 2

3. 3

4. 4

Q.34 In eukaryotes, the regulation could be exerted at -

(A) Transcriptional level (formation of primary transcript)

(B) Processing level (regulation of splicing)

(C) Transport of mRNA from nucleus to the cytoplasm

(D) Translational level

1. (A), (B) and (D) only

2. (A), (B) and (C) only

3. (A), (B), (C) and (D)

4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.35 Arrange the following geological periods in chronological order.

(A). Cretaceous

(B). Triassic

(C). Permian

(D). Jurassic

Choose the correct answer from the options given below:

1. (A), (B), (C), (D)

2. (A), (C), (B), (D)

3. (B), (A), (D), (C)

4. (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.36 If the sequence of bases on one 'sticky end' formed by the action of *EcoRI* is AATT on a Vector DNA, the sequence of bases on 'sticky end' of foreign DNA to which these can join, would be:

1. TATA

2. TTAA

3. AATT

4. UUAA

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Biogas is mixture of:

1. Hydrogen sulphide, nitrogen and methane
2. Hydrogen sulphide, hydrogen and methane
3. Hydrogen sulphide, carbon dioxide and hydrogen
4. Methane, hydrogen sulphide and carbon dioxide.

Options 1. 1

2. 2
3. 3
4. 4

Q.38 Arrange the given steps involved in Polymerase Chain Reaction.

- (A) Annealing
- (B) Extension
- (C) Denaturation
- (D) Amplification

Choose the correct answer from the options given below:

1. (C), (A), (B), (D)
2. (A), (B), (C), (D)
3. (B), (A), (D), (C)
4. (A), (C), (D), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Which one of the following equation represent Verhulst Pearl Logistic Growth?

1. $d(N)/dt = rN (r-N/ K)$
2. $d(N)/dt = rN (K-r/ K)$
3. $d(N)/dt = rN (K-N/ K)$
4. $d(N)/dt = rK (K-N/ K)$

Options 1. 1

2. 2
3. 3
4. 4

Q.40 Myotonic dystrophy is an/a -

1. Autosomal recessive trait
2. Autosomal dominant trait
3. Sex-linked recessive disorder
4. Sex-linked dominant disorder

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the following passage carefully and answer the given questions.

The B-lymphocytes produce an army of proteins in response to pathogens. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection. Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Lymphoid organs are the organs where origin and/or maturation and proliferation of lymphocytes occur. The primary lymphoid organs are bone marrow and thymus. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant *Papaver somniferum*.

Interferons are produced by-

1. Virus infected cells
2. T-lymphocytes
3. B-lymphocytes

4. Bacterial infected cells

Options 1. 1

2. 2

3. 3

4. 4

Q.42 Read the following passage carefully and answer the given questions.
The B-lymphocytes produce an army of proteins in response to pathogens. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection. Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Lymphoid organs are the organs where origin and/or maturation and proliferation of lymphocytes occur. The primary lymphoid organs are bone marrow and thymus. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant *Papaver somniferum*.

Heroin, commonly called smack and chemically is -

1. Acetylcocaine
2. Diacetylmorphine
3. Benzodiazepines
4. Morphine

Options 1. 1

2. 2

3. 3

4. 4

Q.43 Read the following passage carefully and answer the given questions.
The B-lymphocytes produce an army of proteins in response to pathogens. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection. Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Lymphoid organs are the organs where origin and/or maturation and proliferation of lymphocytes occur. The primary lymphoid organs are bone marrow and thymus. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant *Papaver somniferum*.

Which one of the followings drug does not reduce the symptoms of allergy quickly?

1. Anti-histamine
2. Adrenaline
3. Amphetamines
4. Steroids

Options 1. 1

2. 2

3. 3

4. 4

Q.44 Read the following passage carefully and answer the given questions.
The B-lymphocytes produce an army of proteins in response to pathogens. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection. Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy.

Lymphoid organs are the organs where origin and/or maturation and proliferation of lymphocytes occur. The primary lymphoid organs are bone marrow and thymus. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant *Papaver somniferum*.

Which one of the following organs is not a secondary lymphoid organ in humans?

1. Spleen
2. Tonsils
3. Parotid gland
4. Peyer's patches

Options 1. 1

2. 2
3. 3
4. 4

Q.45 Read the following passage carefully and answer the given questions.

The B-lymphocytes produce an army of proteins in response to pathogens. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection. Symptoms of allergic reactions include sneezing, watery eyes, running nose and difficulty in breathing. The use of drugs like anti-histamine, adrenalin and steroids quickly reduce the symptoms of allergy. Lymphoid organs are the organs where origin and/or maturation and proliferation of lymphocytes occur. The primary lymphoid organs are bone marrow and thymus. After maturation the lymphocytes migrate to secondary lymphoid organs like spleen, lymph nodes, tonsils, Peyer's patches of small intestine and appendix. Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine, which is extracted from the latex of poppy plant *Papaver somniferum*.

Which one of the following is not a feature of an antibody:

1. Each antibody has 4 peptide chains
2. Two small peptide chain in antibody structure are called light chains
3. Two longer peptide chains in the antibody structure are called heavy chains
4. T-lymphocytes produce antibodies in response to pathogens in our blood to fight with them

Options 1. 1

2. 2
3. 3
4. 4

Q.46 Read the following passage carefully and answer the given questions.

Insulin consists of two short polypeptide chains- chain A and chain B, that are linked together by disulphide bridges. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans, coleopterans and dipterans. Bt toxin protein is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals. Transgenic animals that produce useful biological products can be created by the introduction of the portion of DNA (or genes) which codes for a particular product such as human protein (alpha-1-antitrypsin) used to treat emphysema. Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses. (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food.

Transgenic animals that produce useful biological products such as human protein (alpha-1-antitrypsin) used to treat the disorder:

1. Adenosine deaminase (ADA) deficiency
2. Cystic fibrosis
3. Phenylketonuria

4. Emphysema

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Read the following passage carefully and answer the given questions. Insulin consists of two short polypeptide chains- chain A and chain B, that are linked together by disulphide bridges. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans, coleopterans and dipterans. Bt toxin protein is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals. Transgenic animals that produce useful biological products can be created by the introduction of the portion of DNA (or genes) which codes for a particular product such as human protein (alpha-1-antitrypsin) used to treat emphysema. Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses. (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food.

Which one of the following species is not killed by insecticidal protein produced by *Bacillus thuringiensis*?

1. Army worm

2. *Meloidegryne incognitia*

3. Flies

4. Beetles

Options 1. 1

2. 2

3. 3

4. 4

Q.48 Read the following passage carefully and answer the given questions. Insulin consists of two short polypeptide chains- chain A and chain B, that are linked together by disulphide bridges. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans, coleopterans and dipterans. Bt toxin protein is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals. Transgenic animals that produce useful biological products can be created by the introduction of the portion of DNA (or genes) which codes for a particular product such as human protein (alpha-1-antitrypsin) used to treat emphysema. Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses. (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food.

The beetles that infest Bt cotton plant are killed:

1. Due to conversion of protoxins into toxin in acidic pH of the gut of the insects that causes lysis of midgut epithelial cells.

2. Due to conversion of protoxins into toxin in alkaline pH of the gut of the insects that causes lysis of midgut epithelial cells.

3. Due to conversion of toxins into protoxin in alkaline pH of the gut of insects that causes lysis of midgut epithelial cells.

4. Due to conversion of protoxins into toxin in alkaline pH of the gut of the insects that does not cause lysis of midgut epithelial cells.

Options 1. 1

2. 2

3. 3

4. 4

Q.49 Read the following passage carefully and answer the given questions.

Insulin consists of two short polypeptide chains- chain A and chain B, that are linked together by disulphide bridges. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans, coleopterans and dipterans. Bt toxin protein is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals. Transgenic animals that produce useful biological products can be created by the introduction of the portion of DNA (or genes) which codes for a particular product such as human protein (alpha-1-antitrypsin) used to treat emphysema. Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses. (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food.

Identify the incorrect fact about genetic modification.

1. It has made crops more tolerant to abiotic stresses (cold, drought, salt, heat).
2. It has reduced reliance on chemical pesticides (pest-resistant crops).
3. It has enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice.
4. It has helped to increase post harvest losses.

Options 1. 1

2. 2
3. 3
4. 4

Q.50 Read the following passage carefully and answer the given questions.

Insulin consists of two short polypeptide chains- chain A and chain B, that are linked together by disulphide bridges. Some strains of *Bacillus thuringiensis* produce proteins that kill certain insects such as lepidopterans, coleopterans and dipterans. Bt toxin protein is converted into an active form of toxin due to the alkaline pH of the gut which solubilise the crystals. Transgenic animals that produce useful biological products can be created by the introduction of the portion of DNA (or genes) which codes for a particular product such as human protein (alpha-1-antitrypsin) used to treat emphysema. Plants, bacteria, fungi and animals whose genes have been altered by manipulation are called Genetically Modified Organisms (GMO). GM plants have been useful in many ways. Genetic modification has: (i) made crops more tolerant to abiotic stresses. (ii) reduced reliance on chemical pesticides (pest-resistant crops). (iii) helped to reduce post harvest losses. (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil). (v) enhanced nutritional value of food.

Which one of the followings is not true about insulin?

1. The main challenge for production of insulin using rDNA techniques was getting insulin assembled into a mature form.
2. The two polypeptide chains in insuline are linked together by disulphide bridges.
3. C-peptide is not present in mature insulin.
4. Insulin is used in diabetic patient in hypoglycemic condition.

Options 1. 1

2. 2
3. 3
4. 4